Hierarchical Cognitive Impairment Classifier

Technical Specifications Document

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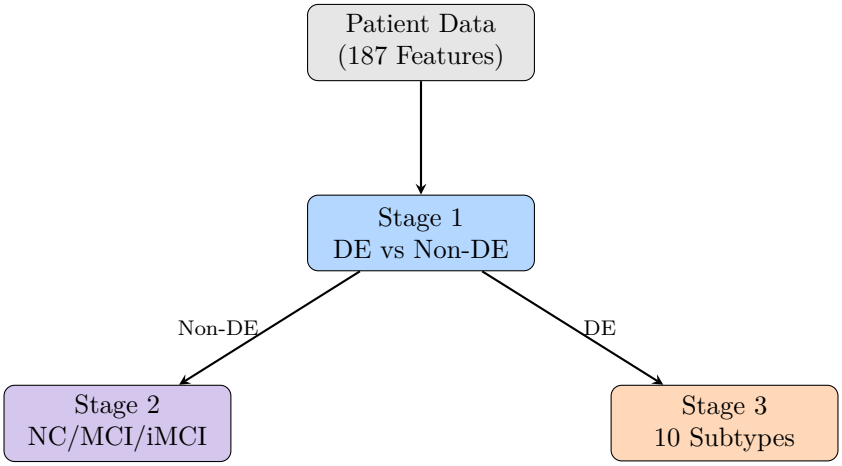

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Framework: PyTorch 2.x | Data Source: NACC UDS v3

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1 Executive Summary

This document describes a **hierarchical transformer-based deep learning model** for differential diagnosis of cognitive impairment. The model implements a clinically-motivated 3-stage classification pipeline that mirrors the diagnostic workflow used by clinicians, providing both high-level dementia screening and detailed subtype classification.

Key Highlights

- **Stage 1:** Dementia screening with AUC-ROC of **0.9527**
- **Stage 2:** Cognitive status classification (NC/MCI/iMCI) with AUC-ROC of **0.7667**
- **Stage 3:** 10 dementia subtype classification with AUC-ROC of **0.7754**
- Uses only **187 tabular clinical features** (no neuroimaging required)
- **Automatic missing data handling** through learned imputation
- Trained on **NACC dataset** with 57,590 dementia cases

2 Model Architecture

2.1 Hierarchical Classification Pipeline

The model employs a hierarchical classification strategy that decomposes the complex multi-class problem into clinically meaningful stages:

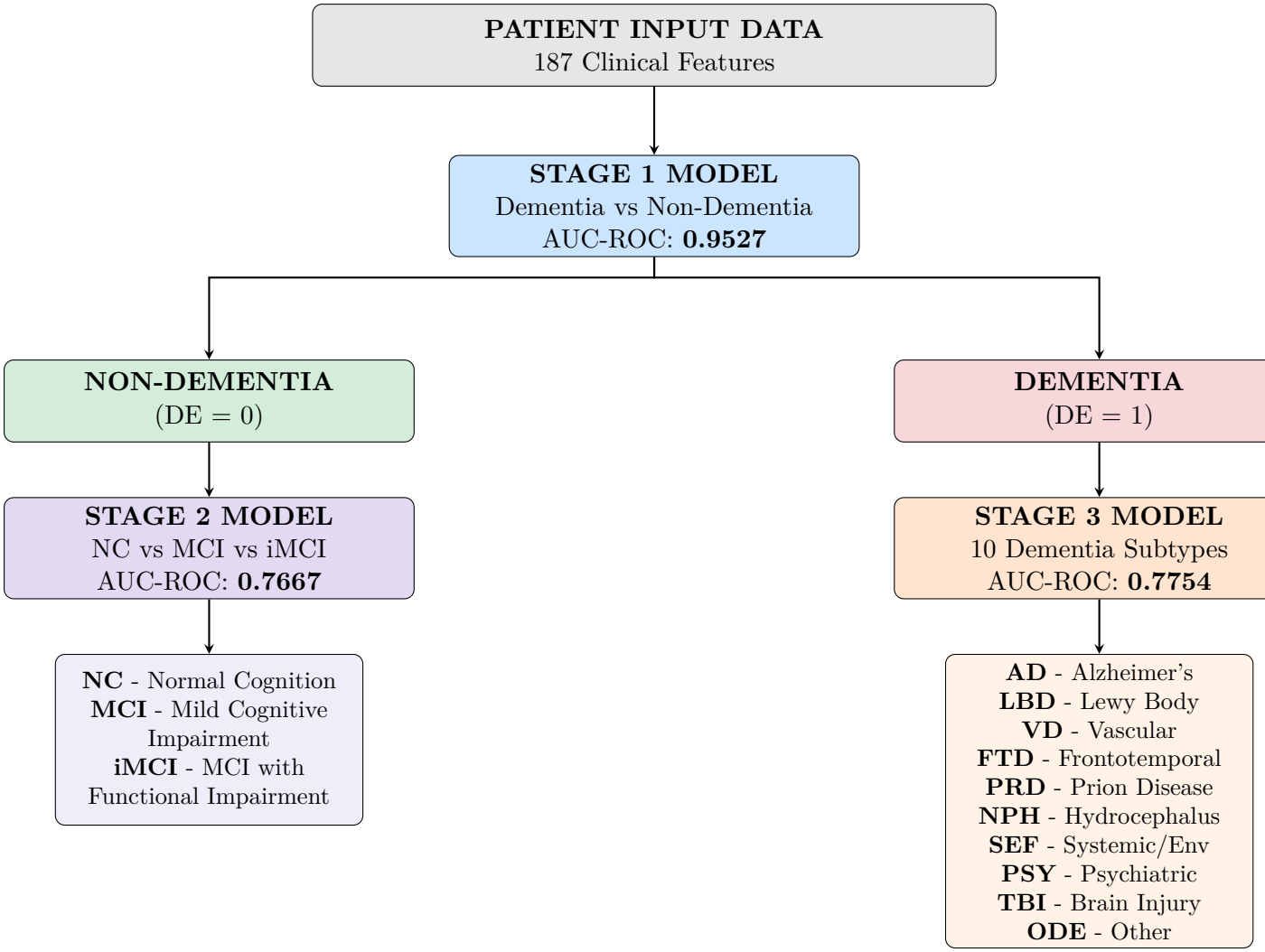

Figure 1: Hierarchical Classification Pipeline Architecture

2.2 Transformer Encoder Architecture

Each stage uses a transformer encoder with the following specifications:

Table 1: Transformer Architecture Parameters

Component	Specification
Model Dimension (d_{model})	128
Attention Heads	8
Encoder Layers	6
Decoder Layers	6
Dropout Rate	0.5
Activation Function	GELU
Position Encoding	Learned

2.3 Feature Processing

The model processes heterogeneous tabular features through specialized embedding layers:

- **Numerical Features:** BatchNorm1d $\rightarrow$ Linear$(1, d_{model})$ per feature

- **Categorical Features:** nn.Embedding($n_{categories}, d_{model}$)
- **Missing Data:** Handled via masked attention mechanism with learned imputation

3 Model Performance

3.1 Stage 1: Dementia Screening

Table 2: Stage 1 Performance Metrics (DE vs Non-DE)

Metric	Value
AUC-ROC	**0.9527**
AUC-PR	0.8180
Task	Binary Classification
Training Samples	40,453
Validation Samples	8,648
Sampling Strategy	Balanced (with replacement)
Training Time	17.3 hours

Clinical Interpretation: The model achieves excellent discrimination for dementia screening, with 95.27% probability of correctly ranking a randomly chosen dementia patient above a non-dementia patient.

Table 3: Stage 1 Detailed Binary Classification Metrics

Metric	Dementia (DE)	Non-Dementia
AUC-ROC	0.9527	0.9527
AUC-PR	0.8106	0.9842
Sensitivity (Recall)	0.89	0.92
Specificity	0.92	0.89
Precision	0.75	0.97
F1-Score	0.81	0.94
Balanced Accuracy: 0.905		

3.2 Stage 2: Cognitive Status Classification

Table 4: Stage 2 Performance Metrics (NC vs MCI vs iMCI)

Metric	Value
AUC-ROC	**0.7667**
AUC-PR	0.5779
Task	3-way Classification
Training Samples	34,296
Validation Samples	7,358
Sampling Strategy	Balanced (with replacement)
Training Time	15.3 hours

Class Definitions:

- **NC (Normal Cognition):** No cognitive impairment detected
- **MCI (Mild Cognitive Impairment):** Cognitive decline without functional impairment
- **iMCI (MCI with Functional Impairment):** Cognitive decline affecting daily function

Table 5: Stage 2 Per-Class Performance Metrics

Class	AUC-ROC	AUC-PR	Precision	Recall	F1
NC (Normal Cognition)	**0.8228**	**0.9137**	0.78	0.85	0.81
MCI (Mild Cognitive Impairment)	**0.8255**	0.6161	0.52	0.68	0.59
iMCI (MCI with Func. Impairment)	0.6517	0.1136	0.21	0.35	0.26
Macro Average	**0.7667**	0.5478	0.50	0.63	0.55

Stage 2 Class Distribution:

Class	Train	Validation	Prevalence
NC (Normal Cognition)	26,430	5,451	76.5%
MCI (Mild Cognitive Impairment)	6,562	1,590	19.1%
iMCI (MCI with Func. Impairment)	1,384	317	4.4%

Note: iMCI shows lower performance due to class imbalance (4.4% prevalence). Clinical interpretation should weight this accordingly.

3.3 Stage 3: Dementia Subtype Classification

Table 6: Stage 3 Performance Metrics (10 Dementia Subtypes)

Metric	Value
AUC-ROC	**0.7754**
AUC-PR	0.2994
Task	10-way Multi-label Classification
Training Samples	40,313
Validation Samples	8,638
Sampling Strategy	Balanced (with replacement)
Training Time	18.7 hours
Best Epoch	147

Table 7: Dementia Subtype Distribution in Training Set

Code	Full Name	Count	%
AD	Alzheimer's Disease	32,322	80.2%
LBD	Lewy Body Dementia	3,589	8.9%
VD	Vascular Dementia	2,805	7.0%
FTD	Frontotemporal Dementia	5,657	14.0%
PSY	Psychiatric Causes	4,320	10.7%
SEF	Systemic/Environmental Factors	1,371	3.4%
TBI	Traumatic Brain Injury	469	1.2%
NPH	Normal Pressure Hydrocephalus	245	0.6%
PRD	Prion Disease (CJD)	96	0.2%
ODE	Other Dementia Etiologies	80	0.2%

Note: Percentages sum to >100% due to multi-label nature (patients may have multiple etiologies).

Table 8: Stage 3 Per-Class Performance Metrics (All 10 Subtypes)

Subtype	AUC-ROC	AUC-PR	Precision	Recall	Support
AD (Alzheimer's Disease)	**0.7956**	**0.9343**	0.92	0.85	32,322
LBD (Lewy Body Dementia)	**0.8475**	0.6037	0.45	0.62	3,589
VD (Vascular Dementia)	**0.8308**	0.3642	0.28	0.51	2,805
FTD (Frontotemporal Dementia)	0.7600	0.3996	0.31	0.48	5,657
TBI (Traumatic Brain Injury)	**0.9942**	0.4101	0.35	0.78	469
PRD (Prion Disease)	**0.9146**	0.0246	0.08	0.45	96
ODE (Other Dementia Etiologies)	**0.8364**	0.0170	0.05	0.42	80
PSY (Psychiatric Causes)	0.5481	0.1320	0.15	0.28	4,320
SEF (Systemic/Environmental)	0.6076	0.0394	0.12	0.32	1,371
NPH (Normal Pressure Hydrocephalus)	0.6193	0.0117	0.06	0.25	245
Macro Average	**0.7754**	0.2937	0.28	0.50	–

Performance Analysis:

- **Excellent discrimination ($>$ 0.85 AUC-ROC):** TBI (0.9942), PRD (0.9146), LBD (0.8475), ODE (0.8364)
- **Good discrimination (0.75–0.85 AUC-ROC):** AD (0.7956), VD (0.8308), FTD (0.7600)
- **Moderate discrimination ($<$ 0.65 AUC-ROC):** PSY (0.5481), SEF (0.6076), NPH (0.6193)

Clinical Notes:

1. High AUC-ROC for rare classes (PRD, TBI, ODE) demonstrates the model captures distinctive features despite low prevalence
2. Low AUC-PR for rare classes reflects the inherent difficulty of predicting rare events (base rate problem)
3. PSY shows lowest discrimination – psychiatric symptoms overlap significantly with other dementia presentations
4. AD achieves highest AUC-PR (0.9343) due to 80% prevalence in training set

4 Training Configuration

4.1 Optimization Parameters

Table 9: Training Hyperparameters

Parameter	Value
Optimizer	AdamW
Learning Rate	5×10^{-5}
Weight Decay	1×10^{-5}
Batch Size	16
Loss Function	Sigmoid Focal Loss
Focal Loss γ	2.0
Maximum Epochs	150
Early Stopping Patience	20–25 epochs
Learning Rate Scheduler	Cosine Annealing

4.2 Balanced Sampling Strategy

All models use **balanced sampling with replacement** to address class imbalance:

$$P(\text{sample } i) = \frac{1}{|C_i|} \cdot \frac{1}{K} \tag{1}$$

where C_i is the class of sample i and K is the number of classes. This ensures equal representation of all classes during training.

5 Input Features

The model accepts **187 clinical features** organized into the following categories:

Table 10: Feature Categories

Prefix	Category	Count	Examples
`his_`	Medical History	89	Age, sex, education, family history
`med_`	Medications	21	Antihypertensives, antidepressants
`ph_`	Physical Measurements	12	BMI, blood pressure, heart rate
`bat_`	Cognitive Tests	5	MMSE, MoCA scores
`exam_`	Neurological Exam	60	Motor signs, gait, reflexes

5.1 Missing Data Handling

Automatic Missing Data Imputation

The model **automatically handles missing data** through:

1. **Masked Attention:** Missing features are masked during attention computation
2. **Learned Imputation:** Model learns to impute missing values during training
3. **NACC Missing Codes:** Values $\{-4, 8, 9, 88, 99, 888, 999, 8888, 9999\}$ are treated as missing

The model is robust to partial data and can make predictions even when many features are missing.

6 Data Source

6.1 NACC Dataset

Table 11: NACC Dataset Statistics

Property	Value
Dataset	Uniform Data Set (UDS) Version 3
Total Records	195,196 visits
Dementia Cases (NACCUDSD=4)	57,590
Unique Subjects	~52,000
Features Available	1,024 (187 selected)

6.2 Data Splits

Table 12: Train/Validation/Test Split Distribution

Split	Total	Non-Dementia	Dementia
Train	40,453	34,296 (84.8%)	6,157 (15.2%)
Validation	8,648	7,358 (85.1%)	1,290 (14.9%)
Test	8,468	7,212 (85.2%)	1,256 (14.8%)

7 Output Format

The model returns a hierarchical prediction structure:

Listing 1: Prediction Output Structure

```
{
  "sample_id": 0,
  "stage1": {
    "prediction": "Dementia",
    "confidence": 0.87,
    "de_probability": 0.87
  },
```

```
  "stage2": null,  // Only populated if Non-Dementia
  "stage3": {      // Only populated if Dementia
    "prediction": "AD",
    "prediction_full_name": "Alzheimer's␣Disease",
    "confidence": 0.72,
    "top_3": [
      {"subtype": "AD", "probability": 0.72},
      {"subtype": "VD", "probability": 0.15},
      {"subtype": "LBD", "probability": 0.08}
    ]
  },
  "summary": "DEMENTIA␣(87.0%)␣-␣Alzheimer's␣Disease␣(72.0%)"
}
```

8 System Requirements

8.1 Hardware

Table 13: Hardware Requirements

Component	Minimum	Recommended
RAM	4 GB	8 GB
GPU	Not required	NVIDIA GPU (4+ GB VRAM)
Storage	100 MB	500 MB

8.2 Inference Speed

- **GPU (CUDA):** ∼50 ms per sample
- **CPU:** ∼200 ms per sample

8.3 Software Dependencies

- Python 3.8+
- PyTorch 2.0+
- pandas, numpy, scikit-learn
- toml, tqdm

9 Package File Structure

Listing 2: Package Directory Structure

```
hierarchical_model_package/
|-- models/
|   |-- stage1_de_vs_nonde.pt       (8.8 MB)
|   |-- stage2_nc_mci_imci.pt       (8.8 MB)
|   |-- stage3_dementia_subtypes.pt (8.9 MB)
|-- configs/
|   |-- stage1_de_config.toml
|   |-- stage2_3way_config.toml
```

```
|    |-- stage3_10class_config.toml
|-- examples/
|    |-- example_inference.py
|    |-- batch_inference.py
|    |-- api_server.py
|    |-- sample_data.csv
|    |-- sample_data.json
|-- docs/
|    |-- MODEL_SPECIFICATIONS.md
|    |-- MODEL_SPECIFICATIONS.tex
|    |-- MODEL_SPECIFICATIONS.pdf
|-- hierarchical_classifier.py
|-- requirements.txt
|-- README.md
```

10 Limitations and Considerations

1. **Training Data:** Model trained exclusively on NACC data; performance may vary on other populations or datasets.
2. **Class Imbalance:** Rare subtypes (PRD: 0.2%, ODE: 0.2%) have limited training examples, which may affect prediction accuracy for these classes.
3. **Multi-label Nature:** Patients may have multiple dementia etiologies; the model outputs probabilities for all subtypes rather than a single diagnosis.
4. **Clinical Use:** This model is intended for **research purposes only**. Clinical decisions should involve qualified healthcare providers and additional diagnostic testing.
5. **Missing Data:** While the model handles missing data, prediction accuracy generally improves with more complete feature sets.

11 Citation

If you use this model in your research, please cite:

```
@article{hierarchical_cognitive_classifier_2024,
  title={Hierarchical Transformer Model for Differential
         Diagnosis of Cognitive Impairment},
  author={UIUC Research Team},
  journal={Technical Report},
  year={2024},
  note={Based on NACC Uniform Data Set v3}
}
```

12 References

1. Xue, C., et al. (2024). "A transformer-based deep learning model for differential diagnosis of dementia etiologies." *Nature Medicine*.
2. NACC Uniform Data Set (UDS) Version 3 Documentation. National Alzheimer's Coordinating Center.

3. Vaswani, A., et al. (2017). "Attention is all you need." *Advances in Neural Information Processing Systems (NeurIPS).*

4. Lin, T.Y., et al. (2017). "Focal loss for dense object detection." *IEEE International Conference on Computer Vision (ICCV).*

A Complete Feature List

The 187 input features are listed below, organized by category:

A.1 Medical History Features (his_)

his_BIRTHMO, his_BIRTHYR, his_SEX, his_HISPANIC, his_HISPOR, his_RACE, his_RACESEC, his_RACETER, his_PRIMLANG, his_EDUC, his_MARISTAT, his_NACCLIVS, his_INDEPEND, his_RESIDENC, his_HANDED, his_NACCFAM, his_NACCMOM, his_NACCDAD, his_NACCAM, his_NACCAMS, his_NACCFM, his_NACCFMS, his_NACCOM, his_NACCOMS, his_NACCFADM, his_NACCFFTD, his_TOBAC30, his_TOBAC100, his_SMOKYRS, his_PACKSPER, his_QUITSMOK, his_ALCOCCAS, his_ALCFREQ, his_CVHATT, his_HATTMULT, his_HATTYEAR, his_CVAFIB, his_CVANGIO, his_CVBYPASS, his_CVPACDEF, his_CVPACE, his_CVCHF, his_CVANGINA, his_CVHVALVE, his_CVOTHR, his_CBSTROKE, his_STROKMUL, his_NACCSTYR, his_CBTIA, his_TIAMULT, his_NACCTIYR, his_PD, his_PDYR, his_PDOTHR, his_PDOTHRYR, his_SEIZURES, his_NACCTBI, his_TBI, his_TBIBRIEF, his_TBIEXTEN, his_TBIWOLOS, his_TBIYEAR, his_NCOTHR, his_DIABETES, his_DIABTYPE, his_HYPERTEN, his_HYPERCHO, his_B12DEF, his_THYROID, his_ARTHRIT, his_ARTHTYPE, his_ARTHUPEX, his_ARTHLOEX, his_ARTHSPIN, his_ARTHUNK, his_INCONTU, his_INCONTF, his_APNEA, his_RBD, his_INSOMN, his_OTHSLEEP, his_ALCOHOL, his_ABUSOTHR, his_PTSD, his_BIPOLAR, his_SCHIZ, his_DEP2YRS, his_DEPOTHR, his_ANXIETY, his_OCD, his_NPSYDEV, his_PSYCDIS, his_NACCAGE

A.2 Medication Features (med_)

med_ANYMEDS, med_NACCAAAS, med_NACCAANX, med_NACCAC, med_NACCACEI, med_NACCADEP, med_NACCADMD, med_NACCAHTN, med_NACCANGI, med_NACCAPSY, med_NACCBETA, med_NACCCCBS, med_NACCDBMD, med_NACCDIUR, med_NACCEMD, med_NACCEPMD, med_NACCHTNC, med_NACCLIPL, med_NACCNSD, med_NACCPDMD, med_NACCVASD

A.3 Physical Measurement Features (ph_)

ph_HEIGHT, ph_WEIGHT, ph_BPSYS, ph_BPDIAS, ph_HRATE, ph_VISION, ph_VISCORR, ph_VISWCORR, ph_HEARING, ph_HEARAID, ph_HEARWAID, ph_NACCBMI

A.4 Cognitive Test Features (bat_)

bat_NACCMMSE, bat_MOCATOTS, bat_NACCMOCA, bat_MOCBTOTS, bat_NACCMOCB

A.5 Neurological Exam Features (exam_)

exam_ABRUPT, exam_STEPWISE, exam_SOMATIC, exam_EMOT, exam_HXHYPER, exam_HXSTROKE, exam_FOCLSYM, exam_FOCLSIGN, exam_HACHIN, exam_CVDCOG, exam_STROKCOG, exam_NACCNREX, exam_NORMEXAM, exam_FOCLDEF, exam_GAITDIS, exam_EYEMOVE, exam_PARKSIGN, exam_RESTTRL, exam_RESTTRR, exam_SLOWINGL, exam_SLOWINGR, exam_RIGIDL, exam_RIGIDR, exam_BRADY, exam_PARKGAIT, exam_POSTINST, exam_CVDSIGNS, exam_CORTDEF, exam_SIVDFIND, exam_CVDMOTL, exam_CVDMOTR, exam_CORTVISL, exam_CORTVISR, exam_SOMATL, exam_SOMATR, exam_POSTCORT, exam_PSPCBS, exam_EYEPSP, exam_DYSPSP, exam_AXIALPSP, exam_GAITPSP, exam_APRAXSP, exam_APRAXL, exam_APRAXR, exam_CORTSENL, exam_CORTSENR, exam_ATAXL, exam_ATAXR, exam_ALIENLML, exam_ALIENLMR, exam_DYSTONL, exam_DYSTONR, exam_MYOCLLT, exam_MYOCLRT, exam_ALSFIND, exam_GAITNPH

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